

Cabin Crew Pairing

Network Flow · DDD · Individual Assignment

Ella

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Problem

Objective	Minimise total cost of cabin crew scheduling
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Methods	MICCPP-ACCS — paper approach (Wen et al. 2022) aggregate flow (LP/MIP) DDD with individual assignment

OG



Model

- Crew scheduled individually, not as teams
- Each class (Senior / Junior / Trainee) has its own availability cap
- Controlled crew substitution (CCS) covers shortfalls across classes
- Big-M penalty for extra crew; substitutions penalised ($\mu \ll M$)

Solution — CG + MIP

- Solve LP relaxation of master problem (Simplex)
- Pricing problem generates columns via resource-constrained shortest path
- Iterate until no negative reduced-cost column exists

- Apply MIP on final column set to recover integer solution
- Heuristic — genetic algorithm used for large instances

Network

- Duty-based: flights $\rightarrow$ duties $\rightarrow$ pairings
- One network per crew class (identical structure)
- All pairings start and end at home base
- Feasibility rules enforced during construction

Planning scope

- Weekly horizon, Single base

MODEL 2 : MICCPP-ACCS (Multi-Class + Controlled Crew Substitution)

Solver status : Optimal

MIP gap : 0.00%

— Workforce summary —

Class 0 (Senior):	avail_cap=	5	used=	5	extra=	0	subs=	0
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Class 1 (Junior):	avail_cap=	7	used=	7	extra=	0	subs=	0
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Class 2 (Trainee):	avail_cap=	5	used=	5	extra=	0	subs=	0
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Total assignments (available crew) : 17

Total substitutions (CCS events) : 0

Total extra-crew assignments : 0

No cross-class substitutions required.

MS (with CCS) = 17 (total crew needed across all flights)

TA (total avail) = 17

Sets & Data

$\mathcal{T}$	Day and time
$\mathcal{P}$	Set of all airports
$\mathcal{B} \subseteq \mathcal{P}$	Crew home bases
$\mathcal{S} \subseteq \mathcal{P} \setminus \mathcal{B}$	Satellite airports ($r \leq 250, \geq 3$ daily flights)
$\mathcal{C}$	Set of individual crew members
$\mathcal{C}_b \subseteq \mathcal{C}$	Crew members based at b
$\mathcal{F}$	Set of all flights
$\mathcal{F}^{\text{cov}} \subseteq \mathcal{F}$	Flights requiring crew coverage (planning period only)
$\mathcal{W}$	Ordered solve windows (rolling horizon)

d_f		Duration of flight f in minutes
dist_f		Great-circle distance of flight f
λ_f		Passenger load factor of flight $f \in [0, 1]$
m_f		Required working crew count for flight f
n_b		Crew supply at base $b \in \mathcal{B}$
$n_{\min}$	5	Minimum number of crew at a base
$n_{\max}$	40	Maximum number of crew introduced by noise at a base
$\pi_{b,p}^w$		Crew from base b at airport p at window w start; $\sum_p \pi_{b,p}^w = n_b$

c_a^{work}		Working crew cost on arc $a \in \mathcal{A}^{\text{fl}}$
c^{fl}	100 / min	Cost per minute of flight time worked
c_a^{dh}	20 / min	Deadhead cost on arc $a \in \mathcal{A}^{\text{dh}}$ (base per-diem + opp.cost)
c^{wt}	0.5 / min	Wait cost rate
C^{ov}	500	Flat penalty per overnight stay away from base (wait ≥ 4 h)
C^{unc}	10^7	Penalty per uncovered crew slot (infeasibility proxy)
Δ^{ta}	45 min	Minimum turnaround between flights at same airport

Δ^{rest}	8 h	Minimum rest before next duty period
Δ^{duty}	14 h	Maximum on-duty time per day
Δ^{hb}	48 h	Minimum home break
D^{work}	5 days	Maximum consecutive days on duty
D^{away}	7 days	Return-window: crew must be back at base
$T_{\text{hor,days}}$		Planning horizon length in days
τ_{duty}		Available duty time per crew member in min
Δ_{bucket}	15 min	Initial DDD time-bucket tolerance; arc is a violation if $ t_{v'} - t^{\text{true}} > \Delta_{\text{bucket}}$

For each base $p \in \mathcal{B}$

$$n_p = \max \left(n_{\min}, \left\lceil \left(\frac{\sum_{f: f.\text{orig} = p} m_f \cdot d_f}{\tau_{\text{duty}}} \cdot 1.5 \right) + \text{noise} \right\rceil \right)$$

- Each base-count yields that many distinct CrewMember objects with unique IDs

Network Building

Symbol	Definition
$v_b^{\text{hor},w}$	Horizon node for base b in window w $v_b^{\text{hor},w} = (b, \mathcal{T}^{\text{hor},w})$
$\mathcal{N} \subseteq \mathcal{P} \times \mathbb{Z}_{>0}$	Time-expanded nodes (p, t) at each dep/arr time
$\mathcal{N}^w \subseteq \mathcal{N}$	Nodes active in window w $\mathcal{N}^w = \{(p, t) \in \mathcal{N} \mid t \leq \mathcal{T}^{\text{hor},w}\}$

- Depot & horizon nodes:** $\forall p \in \mathcal{B} \cup \mathcal{S}$, add $(p, 0)$ and $(p, T^{\text{hor},w})$
- Flight nodes:** $\forall f \in \mathcal{F}$, add $(f.\text{orig}, f.\text{dep})$ and $(f.\text{dest}, f.\text{arr})$

Symbol	Definition
$\mathcal{A} = \mathcal{A}^{\text{fl}} \cup \mathcal{A}^{\text{dh}} \cup \mathcal{A}^{\text{wt}}$	Flight, deadhead, and wait arcs
$\mathcal{A}_v^+, \mathcal{A}_v^- \subseteq \mathcal{A}$	Out-arcs and in-arcs of node v
$\mathcal{K}_b$	Epochs of length $D^{\text{away}} \times 1440$ min covering the horizon for base b
$\mathcal{A}_{b,k}^{\text{hb}} \subseteq \mathcal{A}^{\text{wt}}$	Home-wait arcs at base b in epoch k with cumulative home-wait $\geq \Delta^{\text{hb}}$

1. **Wait arcs:** Crew waits at airport
 - $\text{cost} = c^{\text{wt}} \cdot \Delta t + C^{\text{ov}} \cdot \mathbb{1}[\Delta t \geq 4h]$
 - chain consecutive nodes at each airport

2. **Flight arcs:** Crew works the leg
 - $\text{cost} = c^{\text{fl}} \cdot d_f$
 - forward pass in time order
 - snap arrival node
 - check duty limit
 - propagate home-break clocks

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 - $\text{cost} = c^{\text{fl}} \cdot d_f$
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 - snap arrival node
 - check duty limit
 - propagate home-break clocks
3. **Deadhead arcs:** Crew repositions as passenger
 - $\text{cost} = c^{\text{dh}} \cdot d_f + \text{opp_cost}(f)$

Arrival end of a flight arc is not placed at $f.\text{arr}$, but snapped forward to the earliest existing node at $f.\text{dest}$ satisfying the gap

$\mathcal{T}_p = \{t : (p, t) \in \mathcal{N}\}$ is the set of timestamps at airport p

$$v' = (f.\text{dest} : \min\{t' \in \mathcal{T}_{f.\text{dest}} \wedge t' \geq f.\text{arr} + \Delta^{\text{ta}}\})$$

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Working and deadheading crew share the same arc both land at v' .
Depot departures $(p, 0)$ are exempt.

For each base $b \in \mathcal{B}$

$\mathcal{G}^w := (\mathcal{N}^w, \mathcal{A})$ - the window's time-expanded graph

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- Fwd_b :
 - Dijkstra from depot nodes $(p, 0)$, $p \in \mathcal{B} \cup \mathcal{S}$
 - home-break clock state — arcs violating D^{work} or D^{away} are excluded
- Bwd_b : Dijkstra backwards from $(b, T^{\text{hor}, w})$

Rolling Horizon

For window $w \in \mathcal{W}$, with $M = 1440$ min/day:

$$T^{\text{start},w} = (w - 1) \cdot S_{\text{days}} \cdot M$$

$$T^{\text{commit},w} = T^{\text{start},w} + S_{\text{days}} \cdot M$$

$$T^{\text{hor},w} = T^{\text{start},w} + (W_{\text{days}} + R_{\text{days}}) \cdot M$$

W_{days}	7 days	Solve window length
S_{days}	3 days	Days committed per step
$W_{\text{days}} - S_{\text{days}}$	4 days	Look-ahead overlap
R_{days}	7 days	Return tail

Flights in window w :

$$\mathcal{F}^w = \{f \in \mathcal{F} \mid T^{\text{start},w} \leq \text{dep}_f < T^{\text{hor},w}\}$$

Flights needing coverage:

$$\mathcal{F}^{\text{cov},w} = \{f \in \mathcal{F}^w \mid \text{dep}_f < T^{\text{commit},w}\}$$

Let $\rho_c^w \in \mathcal{P}$ be crew member c 's location at $T^{\text{commit},w}$, derived from the path-tracing step as the airport of the last node visited at or before $T^{\text{commit},w}$.

$$\pi_{b,p}^{w+1} = |\{c \in \mathcal{C}_b \mid \rho_c^w = p\}| \quad \forall b \in \mathcal{B},; p \in \mathcal{P}$$

Supply totals validated: $\sum_p \pi_{b,p}^{w+1} = n_b$; rounding absorbed at home depot.

Clock state per crew c : τ_c last reset time (min) d_c work-days since
 τ_c h_c home-wait since last departure.

Worst-case clock state tuple $\Gamma_b^{w+1} = (d_c, \tau_c, h_c)$ carried forward:

$$\Gamma_b^{w+1} = \arg \max_{c \in \mathcal{C}_b} d_c^w$$

where d_c^w, τ_c^w, h_c^w denote the values of the crew-clock state variables at the end of window w .

Ties broken by earliest reset τ_c^w , then lowest home-wait h_c^w .

DDD — Individual Model

- $x_{c,a} \in [0, 1]$: Arc indicator for crew member $c \in \mathcal{C}$ on arc $a \in \mathcal{A}$; relaxed to continuous in LP
- $\sigma_f \in [0, m_f]$: Coverage slack, penalised at C^{unc}

$$\begin{aligned}
 \min \quad & \underbrace{\sum_{c \in \mathcal{C}} \sum_{a \in \mathcal{A}^{\text{fl}}} c^{\text{fl}} \cdot d_{f(a)} \cdot x_{c,a}}_{\text{flight hours}} \\
 & + \underbrace{\sum_{c \in \mathcal{C}} \sum_{a \in \mathcal{A}^{\text{dh}}} c_a^{\text{dh}} \cdot x_{c,a}}_{\text{deadhead}} \\
 & + \underbrace{\sum_{c \in \mathcal{C}} \sum_{a \in \mathcal{A}^{\text{wt}}} (c^{\text{wt}} \cdot \Delta t(a) + C^{\text{ov}} \cdot \mathbb{1}[\Delta t(a) \geq 4h]) \cdot x_{c,a}}_{\text{layover / wait}} \\
 & + \underbrace{\sum_{f \in \mathcal{F}^{\text{cov}}} C^{\text{unc}} \cdot \sigma_f}_{\text{uncovered slots}}
 \end{aligned}$$

Crew **cannot** start from or end at another base's depot/horizon node.

For each crew member c with home base b and each node $v \in \mathcal{N}$:

$$\sum_{a \in \mathcal{A}_v^+} x_{c,a} - \sum_{a \in \mathcal{A}_v^-} x_{c,a} = \begin{cases} +1 & v = (b, 0) \quad \text{home depot} \\ -1 & v = (b, T^{\text{hor},w}) \quad \text{home horizon} \\ 0 & \text{otherwise} \end{cases}$$

Constraint 2': Flight Coverage

Each flight must be staffed to m_f working crew, or pay C^{unc} per missing slot.

Let $\mathcal{A}_f = \{a \in \mathcal{A}^{\text{fl}} \mid a.\text{flight} = f\}$ be the working arcs for flight f .

$$\sum_{c \in \mathcal{C}} \sum_{a \in \mathcal{A}_f} x_{c,a} + \sigma_f \geq m_f \quad \forall f \in \mathcal{F}^{\text{cov}}$$

Constraint 3': Duty Time & Home Break DDD – Individual Model (Structural)

Both enforced during **arc construction** — no MIP constraints needed.

Duty time: $d^{\text{duty}}(v) + d_f > \Delta^{\text{duty}} \implies$ arc not created.

Rest arc ($\text{wait} \geq \Delta^{\text{rest}}$) resets: $d^{\text{duty}}(v') \leftarrow 0$.

Home break & return window: arcs violating D^{work} or D^{away} excluded during reachability pruning (Fwd pass carries clock state).

Constraint 4': Non-Negativity & Integrality

$$x_{c,a} \in (0, 1) \quad \forall c \in \mathcal{C}, a \in \mathcal{A}$$

$$\sigma_f \in \mathbb{Z}_{\geq 0} \quad \forall f \in \mathcal{F}^{\text{cov}}$$

DDD iteratively refines the network till LP **arc-feasible**

1. **Solve LP** on current network

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1. **Solve LP** on current network

2. **Detect violations:**

- node too coarse to represent true arrival

$$|t_{v'} - t^{\text{true}}| > \Delta_{\text{bucket}}$$

- turnaround infeasible in current discretisation

$$t^{\text{true}} + \Delta^{\text{ta}} > t_{\text{next dep}}$$

- capped at 500 per iteration, sorted by airport activity

3. Refine:

- insert (p, t^*) at each violated time
- bisect spanning wait arc
- expose new flight/deadhead arcs
- update flow balance and coverage in-place

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- insert (p, t^*) at each violated time
- bisect spanning wait arc
- expose new flight/deadhead arcs
- update flow balance and coverage in-place

4. **Repeat** until no violations

5. **Switch to MIP:**

- $x_{c,a}$ to binary, σ_f to integer
- `MIPGap = 0.01, TimeLimit = 300s`

For each crew member c with active arcs $\mathcal{A}_c^*$:

1. Build adjacency $\text{out}(v) = a$ for each node (prefer flight > deadhead > wait)
2. Trace path from home depot $(b, 0)$ following $\text{out}(\cdot)$
3. Emit each non-wait leg as a route entry: type, origin, destination, departure, arrival

$$\text{Route}(c) = \{\ell \in \mathcal{A}_c^* \mid \ell.\text{type} \neq \text{wait}\}$$

Each route entry:

<code>crew_id</code>	Unique crew member identifier
<code>base</code>	Home base airport code
<code>legs</code>	Ordered list of leg dictionaries

Each leg:

<code>type</code>	"flight" or "deadhead"
<code>from, to</code>	Airport codes
<code>dep, arr</code>	Minutes from week start
<code>flight_id</code>	Reference to the covered flight (or null)

```
=====
ROLLING WINDOW 1 | days 1-3 (solve horizon: day 14)
=====
```

Building network...

Depot/horizon nodes: 20 (0.0s)

All nodes added: 13804 (0.0s)

Wait arcs built (0.0s)

WARNING: 39 flights have no connectable arc!

Turnaround-pruned: 18575 arcs (< 45 min ground time)

Duty-pruned : 11938 arcs (exceeded 20h duty / 5d period since last 8h rest)

Flight arcs: 6637 (0.2s)

Network complete: 13804 nodes, 20431 arcs (total 0.2s)

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Building flow model: 20431 arcs, 13804 nodes, 10 bases (0.0s)

Computing reachability per base (parallel Dijkstra)... done (0.5s)

Reachable (base,arc) pairs: 114,556 / 204,310 = 56.1% of full cross-product

Variables: 19,348,099 (incl. 15,574,980 per-crew flight, 3,608,811 per-crew home-wait) (40.1s)

Flow balance constraints: 138,040 (46.1s)

Linking constraints (var_work_crew==var_work): 59,481 (54.8s)

Per-crew home-balance constraints: 3,611,433 (84.1s)

Aggregate Flow Model

$f_{b,a}^{\text{work}} \in \mathbb{Z}_{\geq 0}$	Working crew from base b on arc $a \in \mathcal{A}^{\text{fl}}$
$f_{b,a}^{\text{dh}} \in \mathbb{Z}_{\geq 0}$	Deadheading crew from base b on arc $a \in \mathcal{A}^{\text{dh}}$
$f_{b,a}^{\text{wt}} \in \mathbb{Z}_{\geq 0}$	Waiting crew from base b on arc $a \in \mathcal{A}^{\text{wt}}$
$\varphi_{b,a}$	$f_{b,a}^{\text{work}} + f_{b,a}^{\text{dh}} + f_{b,a}^{\text{wt}}$ – total flow from base b on arc a
$\sigma_f \in \mathbb{Z}_{\geq 0}$	Uncovered crew slots for flight f

$$\begin{aligned}
 \min \quad & \underbrace{\sum_{b,a \in \mathcal{A}^{\text{fl}}} c_a^{\text{work}} f_{b,a}^{\text{work}}}_{\text{flight hours}} \\
 & + \underbrace{\sum_{b,a \in \mathcal{A}^{\text{dh}}} c_a^{\text{dh}} f_{b,a}^{\text{dh}}}_{\text{deadhead}} \\
 & + \underbrace{\sum_{b,a \in \mathcal{A}^{\text{wt}}} c_a^{\text{wt}} f_{b,a}^{\text{wt}}}_{\text{layover / wait}} \\
 & + \underbrace{\sum_{f \in \mathcal{F}^{\text{cov}}} C^{\text{unc}} \cdot \sigma_f}_{\text{uncovered slots}}
 \end{aligned}$$

Flow conserved at every interior node; supply injected at depots.

$$\sum_{a \in \mathcal{A}_v^+} \varphi_{b,a} - \sum_{a \in \mathcal{A}_v^-} \varphi_{b,a} = \begin{cases} \pi_{b,p}^w & v = (p, 0) \\ 0 & \text{otherwise} \end{cases} \quad \forall p \in \mathcal{P}$$

$$\forall b \in \mathcal{B}, \forall v \in \mathcal{N}^w \setminus \{v_b^{\text{hor},w}\}$$

Final window: all crew return home. Intermediate: free-exit sink.

$$\sum_{a \in \mathcal{A}_{v_b}^{-}, w} \varphi_{b,a} - \sum_{a \in \mathcal{A}_{v_b}^{+}, w} \varphi_{b,a} = \begin{cases} n_b & \text{final window } (w = |\mathcal{W}|) \\ \geq 0 & \text{intermediate window} \end{cases}$$

Each flight must be staffed to m_f working crew, or pay C^{unc} per missing slot.

$$\sum_{b \in \mathcal{B}} \sum_{a \in \mathcal{A}_f} f_{b,a}^{\text{work}} + \sigma_f \geq m_f \quad \forall f \in \mathcal{F}^{\text{cov}}$$

Where $\mathcal{A}_f = \{a \in \mathcal{A}^{\text{fl}} \mid a.\text{flight} = f\}$ be the working arcs for flight f

Enforced by **arc construction** — no MIP row added.

If accumulated duty at node v would exceed the limit on flight f , the arc is never created:

$$d^{\text{duty}}(v) + d_f > \Delta^{\text{duty}} \implies \text{arc not created}$$

Propagated forward through the network:

$$d^{\text{duty}}(v') = \min(d^{\text{duty}}(v'), d^{\text{duty}}(v) + d_f)$$

A rest arc ($\text{wait} \geq \Delta^{\text{rest}}$) resets: $d^{\text{duty}}(v') \leftarrow 0$.

Enforced by **both** arc pruning and an explicit MIP constraint.

Structural: Any arc that would be taken while $d > D^{\text{work}}$ or $\frac{t-\tau}{1440} > D^{\text{away}}$ is pruned

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Structural: Any arc that would be taken while $d > D^{\text{work}}$ or $\frac{t-\tau}{1440} > D^{\text{away}}$ is pruned

MIP constraint: for each base b and each epoch $k \in \mathcal{K}_b$ of length D^{away} , total flow returning home must cover the full crew

$$\sum_{a \in \mathcal{A}_{b,k}^{\text{hb}}} f_{b,a}^{\text{wt}} \geq n_b \quad \forall b \in \mathcal{B}, k \in \mathcal{K}_b$$

Constraint 7: Non-Negativity & Integrality

Aggregate Flow Model

$$f_{b,a}^{\text{work}}, f_{b,a}^{\text{dh}}, f_{b,a}^{\text{wt}} \in \mathbb{Z}_{\geq 0} \quad \forall b \in \mathcal{B}, a \in \mathcal{A}$$

$$\sigma_f \in \mathbb{Z}_{\geq 0} \quad \forall f \in \mathcal{F}^{\text{cov}}$$

```
=====
ROLLING WINDOW 1 | days 1-3 (solve horizon: day 14)
=====
```

Building network...

Depot/horizon nodes: 20 (0.0s)

All nodes added: 13804 (0.0s)

Wait arcs built (0.1s)

WARNING: 39 flights have no connectable arc!

Turnaround-pruned: 18575 arcs (< 45 min ground time)

Duty-pruned : 11938 arcs (exceeded 20h duty / 5d period since last 8h rest)

Flight arcs: 6637 (0.2s)

Network complete: 13804 nodes, 20431 arcs (total 0.2s)

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Building flow model: 20431 arcs, 13804 nodes, 10 bases (0.0s)

Computing reachability per base (parallel Dijkstra)... done (0.6s)

Reachable (base,arc) pairs: 114,556 / 204,310 = 56.1% of full cross-product

Variables: 178,065 (1.0s)

Flow balance constraints: 138,040 (1.9s)

Coverage constraints: 4028 (2.0s)

Mandatory home-break constraints: 20 (2.0s)

Model built: 178,065 vars, 142,088 constrs (2.0s)

```
=====
SOLUTION SUMMARY
=====
Mode           : Rolling horizon
Status         : rolling_horizon
Total cost     : 2,489,313,613.4
  Flight hours : 2,444,027,600.0
  Deadhead     : 25,176,100.0
  Layover/wait : 20,109,913.4
Flights        : 40579
Covered        : 40579
Uncovered slots : 0.0
Individual routes: 2554
Solve time     : 385.6s
```

```
=====
ROLLING WINDOW 1 | days 1-3 (solve horizon: day 14)
=====
```

Building network...

Depot/horizon nodes: 388 (0.0s)

All nodes added: 341447 (1.3s)

Wait arcs built (3.8s)

WARNING: 458 flights have no connectable arc!

Turnaround-pruned: 246036 arcs (< 45 min ground time)

Duty-pruned : 144079 arcs (exceeded 20h duty / 5d period since last 8h rest)

HB-pruned : 36 arcs fully pruned (all bases exceeded MAX_WORK_DAY=5 or MAX_AWAY_DAYS=7)

Flight arcs: 101921 (96.9s)

Network complete: 341447 nodes, 412355 arcs (total 96.9s)

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Building flow model: 412355 arcs, 341447 nodes, 194 bases (0.0s)

Computing reachability per base (parallel Dijkstra)... done (28.8s)

Reachable (base,arc) pairs: 34,855,562 / 79,996,870 = 43.6% of full cross-product

Variables: 49,750,734 (343.8s)

Flow balance constraints: 66,240,718 (1277.8s)

Coverage constraints: 57854 (1298.1s)

Mandatory home-break constraints: 388 (1305.2s)

Model built: 49,750,734 vars, 66,298,960 constrs (1305.2s)

Stage 2: Flow Decomposition

For each base b :

1. Build residual: $\text{res}(b, a) = f_{b,a}^{\text{work}} + f_{b,a}^{\text{dh}} + f_{b,a}^{\text{wt}}$
2. For each depot group (non-home first, home last):
 - BFS to find arcs reachable from this depot
 - Trace one path per crew member; consume arcs from residual
3. Classify each leg: *working* if $(b, a) \in \text{arc_is_work}$, else *deadheading*

BFS state: (v, g) where $g = 1$ if home-wait $h_c \geq \Delta^{\text{hb}}$ (gate open).

Home-break gate: at home base with $g = 0$, only wait or return-to-home arcs are traversable.

BFS state: (v, g) where $g = 1$ if home-wait $h_c \geq \Delta^{\text{hb}}$ (gate open).

Home-break gate: at home base with $g = 0$, only wait or return-to-home arcs are traversable.

Home-wait accumulation

$$h_{c'} = \begin{cases} h_c + (t_{\text{end}} - t_{\text{start}}) & \text{wait arc at home base} \\ 0 & \text{flight/dh arrives at home base} \\ h_c & \text{otherwise} \end{cases}$$

Arc priority: flight > deadhead > wait.

Let $\text{legs}^w(c)$ denote the set of legs (non-wait arcs) assigned to crew member c in window w , obtained from path tracing.

$$\text{Route}(c) = \bigsqcup_{w \in \mathcal{W}} \{ \ell \in \text{legs}^w(c) \mid \text{dep}(\ell) < T^{\text{commit}, w} \}$$

Legs sorted chronologically; overlap duplicates removed by (from, to, dep, arr) key.

After path tracing, simulate clocks per crew c :

- τ_c : last clock reset (min)
- d_c : work-days since τ_c
- h_c : home-wait since last departure

Reset on departure from home with $h_c \geq \Delta^{\text{hb}}$:

$$\tau_c \leftarrow \text{dep} \quad d_c \leftarrow 0 \quad h_c \leftarrow 0$$

Violation flags (first breach only):

$$\text{Away} : \frac{f.\text{dep} - \tau_c}{1440} > D^{\text{away}} \quad \text{Work} : d_c > D^{\text{work}}$$

Worst-case state forwarded to next window via Γ_b^{w+1} .

```
=====
ROLLING WINDOW 2 | days 4–6 (solve horizon: day 17)
=====
```

...

WARNING: crew 69 (base ATL) fallback to home – position gap: was at BOS, no flow found there;
starting from ATL instead.

WARNING: crew 1929 (base ORD) fallback to home – position gap: was at SEA, no flow found there;
starting from ORD instead.

WARNING: crew 1931 (base ORD) fallback to home – position gap: was at SEA, no flow found there;
starting from ORD instead.

WARNING: crew 2166 (base SEA) fallback to home – position gap: was at MIA, no flow found there;
starting from SEA instead.

WARNING: crew 2157 (base SEA) fallback to home – position gap: was at LAX, no flow found there;
starting from SEA instead.

Window status : optimal

Window cost : 240,588,185.0

Flights in window : 4028

Covered : 4028

Uncovered slots : 0.0

Home-break violations : 97 (crew rescheduling needed)

=====

ROLLING WINDOW 10 | days 28–30 (solve horizon: day 37)

=====

HOME-BREAK violation crew 2 (ATL): Day 28: worked 7d > 5d (BOS->ORD)

WARNING: crew 105 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 106 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 109 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 116 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 129 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 132 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 133 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 134 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 139 (base ATL) fallback to home – position gap: was at DFW, no flow found there; starting from ATL instead.

WARNING: crew 141 (base ATL) fallback to home – position gap: was at DFW, no flow found there;

starting from ATL instead.

WARNING: crew 149 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 153 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 154 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 156 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 161 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 162 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 168 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 171 (base ATL) fallback to home – position gap: was at JFK, no flow found there;
starting from ATL instead.

WARNING: crew 179 (base ATL) fallback to home – position gap: was at JFK, no flow found there;
starting from ATL instead.

WARNING: crew 187 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 188 (base ATL) fallback to home – position gap: was at DFW, no flow found there;
starting from ATL instead.

WARNING: crew 96 (base ATL) fallback to home – position gap: was at SF0, no flow found there; starting from ATL instead.

WARNING: crew 97 (base ATL) fallback to home – position gap: was at SF0, no flow found there; starting from ATL instead.

WARNING: crew 98 (base ATL) fallback to home – position gap: was at SF0, no flow found there; starting from ATL instead.

WARNING: crew 102 (base ATL) fallback to home – position gap: was at SF0, no flow found there; starting from ATL instead.

WARNING: crew 183 (base ATL) fallback to home – position gap: was at SF0, no flow found there; starting from ATL instead.

WARNING: crew 184 (base ATL) fallback to home – position gap: was at SF0, no flow found there; starting from ATL instead.

...

Window status : optimal

Window cost : 250,759,659.1

Flights in window : 4111

Covered : 4111

Uncovered slots : 0.0

Home-break violations : 1962 (crew rescheduling needed)

Demo

Complexity Comparison

N	$= \mathcal{N} $	nodes
A	$= \mathcal{A} $	arcs
A^{reach}	$\subseteq A$	arcs surviving reachability pruning (typically $A^{\text{reach}} \ll C \cdot A$)
F	$= \mathcal{F}^{\text{cov}} $	covered flights
B	$= \mathcal{B} $	bases
C	$= \mathcal{C} $	crew
W	$= \mathcal{W} $	windows

	Aggregate Flow	DDD – Individual
Vars	$3BA + F$ integer	$CA^{\text{reach}} + F$ binary
Flow balance	$B(N - B)$	CN
Coverage	FW	FW
Home-break	$B \ \mathcal{K} $	<i>structural</i>
MIP	$O(BA)$	$O(CA^{\text{reach}})$
Extraction	$O(BA)$ BFS	$O(CA^{\text{reach}})$
DDD overhead	—	$O(K_{\text{iter}}(A + CN_{\text{new}}))$

Future Work

Existing Models

- Extract the schedule correctly
- Senior, Junior substitutions

New Model

- Figure out a way to solve for US Domestic 2025/01 Dataset (599k Entries) in reasonable time running on my mac